

SEONGKU KANG

Postdoctoral Researcher, Computer Science, University of Illinois at Urbana-Champaign (UIUC)

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RESEARCH INTERESTS

- Data mining and machine learning for real-world applications
- Information retrieval and recommender systems

EDUCATION

Ph.D., Computer Science and Engineering, POSTECH Mar 2018 - Aug 2023

- Dissertation: Knowledge Distillation Approaches for Accurate and Efficient Recommender System

B.S., Computer Science and Engineering, NTU Jul 2016 - May 2017

- Exchange Student

B.S., Computer Science and Engineering, Hanyang University Mar 2012 - Feb 2018

- Cumulative GPA: 4.43/4.50 (ranked 1st in CSE Dept.).

SELECTED PUBLICATIONS

[**KDD'24**] Gyuseok Lee*, SeongKu Kang*, Wonbin Kweon, Hwanjo Yu (co-first authors)
Continual Collaborative Distillation for Recommender System

[**WWW'24**] SeongKu Kang, Shivam Agarwal, Bowen Jin, Dongha Lee, Hwanjo Yu, Jiawei Han
Improving Retrieval in Theme-specific Applications using a Corpus Topical Taxonomy

[**WWW'23**] SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu
Distillation from Heterogeneous Models for Top-K Recommendation

[**CIKM'23**] Youngjune Lee, Keunchan Park, Yeongjong Jeong, SeongKu Kang (corresponding author)
MvFS: Multi-view Feature Selection for Recommender System

[**WWW'22**] SeongKu Kang, Dongha Lee, Wonbin Kweon, Junyoung Hwang, Hwanjo Yu
Consensus Learning from Heterogeneous Objectives for One-Class Collaborative Filtering

[**KDD'21**] SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
Topology Distillation for Recommender System

[**CIKM'20**] SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
DE-RRD: A Knowledge Distillation Framework for Recommender System

[**CIKM'19**] SeongKu Kang, Junyoung Hwang, Dongha Lee, Hwanjo Yu
Semi-Supervised Learning for Cross-Domain Recommendation to Cold-start Users

EXPERIENCE

Data Mining Group, UIUC Jun 2023 - Present

- Postdoctoral Researcher (advisor: [Prof. Jiawei Han](#))

Microsoft Research Asia, Beijing, China Oct 2022 - April 2023

- Research intern in Social Computing group (mentors: [Dr. Xing Xie](#), [Dr. Jianxun Lian](#))
- Award of excellence (Stars of Tomorrow, top 10%)

AWARDS & FELLOWSHIPS

Awards

- Stars of Tomorrow Award for Outstanding Intern, Microsoft Research Asia, 2023
- POSTECH Research Award (4th place), National Research Foundation, Korea, 2022
- Music Playlist Recommendation Competition (3rd place), Kakao, 2020
- SIGIR Student Travel Award, ACM SIGIR, 2019-2020
- Dean's List, Hanyang University, 2018
- Best Paper Award (3rd place), Korean Database Conference (KDBC), 2017

Fellowships

- POSTECHIAN Fellowship, POSTECH, 2023
- NAVER Ph.D Fellowship, NAVER, 2021
- NAVER Academic Scholarship, NAVER, 2017
- National Science and Technology Scholarship, Ministry of Education, Korea, 2012-2017
- Hanyang Brain Scholarship, Hanyang University, 2012

TALKS

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| • CS/AI Seminar, Ulsan National Institute of Science and Technology (UNIST) | Nov 2023 |
| • Invited Seminar, Electronic & Information Research Information Center (EIRIC) | Sep 2023 |
| • Invited Seminar, Social Computing Group, Microsoft Research Asia (MSRA) | Jun 2023 |
| • Technology Showcase, Korea AI summit | Dec 2022 |

FULL PUBLICATIONS

Unless otherwise specified, the paper is accepted as a long/regular paper. * indicates equal contribution.

Conferences

1. Continual Collaborative Distillation for Recommender System
Gyuseok Lee*, SeongKu Kang*, Wonbin Kweon, Hwanjo Yu
KDD'24 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
2. PEARL: A Review-driven Persona-Knowledge grounded Conversational Recommendation Dataset
Minjin Kim*, Minju Kim*, Hana Kim, Beong-woo Kwak, Soyeon Jeon, Hyunseo Kim, SeongKu Kang,
Youngjae Yu, Jinyoung Yeo, Dongha Lee
ACL'24 Findings | Annual Meeting of the Association for Computational Linguistics
3. Self-Consistent Reasoning-based Aspect-Sentiment Quad Prediction with Extract-Then-Assign Strategy
Jieyong Kim*, Ryang Heo*, Yongsik Seo, SeongKu Kang, Jinyoung Yeo, Dongha Lee
ACL'24-short Findings | Annual Meeting of the Association for Computational Linguistics
4. Multi-Domain Sequential Recommendation via Domain Space Learning
Junyoung Hwang, Hyunjun Ju, SeongKu Kang, Sanghwan Jang, Hwanjo Yu
SIGIR'24 | ACM SIGIR Conference on Research and Development in Information Retrieval
5. Learning Discriminative Dynamics with Label Corruption for Noisy Label Detection
Suyeon Kim, Dongha Lee, SeongKu Kang, Sukang Chae, Sanghwan Jang, Hwanjo Yu
CVPR'24 | IEEE / CVF Computer Vision and Pattern Recognition
6. Improving Retrieval in Theme-specific Applications using a Corpus Topical Taxonomy
SeongKu Kang, Shivam Agarwal, Bowen Jin, Dongha Lee, Hwanjo Yu, Jiawei Han
WWW'24 | ACM The Web Conference

7. Top-Personalized-K Recommendation
Wonbin Kweon, SeongKu Kang, Sanghwan Jang, Hwanjo Yu
WWW'24 | ACM The Web Conference
8. Multi-Domain Recommendation to Attract Users via Domain Preference Modeling
Hyunjun Ju, SeongKu Kang, Dongha Lee, Junyoung Hwang, Sanghwan Jang, Hwanjo Yu
AAAI'24 | AAAI Conference on Artificial Intelligence
9. MvFS: Multi-view Feature Selection for Recommender System
Youngjune Lee, Keunchan Park, Yeongjong Jeong, SeongKu Kang (corresponding author)
CIKM'23-short | ACM International Conference on Information and Knowledge Management
10. Distillation from Heterogeneous Models for Top-K Recommendation
SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu
WWW'23 | ACM The Web Conference
11. Learning Topology-Specific Experts for Molecular Property Prediction
Suyeon Kim, Dongha Lee, SeongKu Kang, Seonghyeon Lee, Hwanjo Yu
AAAI'23 | AAAI Conference on Artificial Intelligence
12. Consensus Learning from Heterogeneous Objectives for One-Class Collaborative Filtering
SeongKu Kang, Dongha Lee, Wonbin Kweon, Junyoung Hwang, Hwanjo Yu
WWW'22 | ACM The Web Conference
13. TaxoCom: Topic Taxonomy Completion with Hierarchical Discovery of Novel Topic Clusters
Dongha Lee, Jiaming Shen, SeongKu Kang, Susik Yoon, Jiawei Han, Hwanjo Yu
WWW'22 | ACM The Web Conference
14. Obtaining Calibrated Probabilities with Personalized Ranking Models
Wonbin Kweon, SeongKu Kang, Hwanjo Yu
AAAI'22 | AAAI Conference on Artificial Intelligence
15. Topology Distillation for Recommender System
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
KDD'21 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
16. Bootstrapping User and Item Representations for One-Class Collaborative Filtering
Dongha Lee, SeongKu Kang, Hyunjun Ju, Chanyoung Park, Hwanjo Yu
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval
17. Unsupervised Proxy Selection for Session-based Recommender Systems
Junsu Cho, SeongKu Kang, Dongmin Hyun, Hwanjo Yu
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval
18. Learning Heterogeneous Temporal Patterns of User Preference for Timely Recommendation
Junsu Cho, Dongmin Hyun, SeongKu Kang, Hwanjo Yu
WWW'21 | ACM International World-Wide Web Conference
19. Bidirectional Distillation for Top-K Recommender System
Wonbin Kweon, SeongKu Kang, Hwanjo Yu
WWW'21 | ACM International World-Wide Web Conference
20. DE-RRD: A Knowledge Distillation Framework for Recommender System
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
CIKM'20 | ACM International Conference on Information and Knowledge Management

21. Deep Rating Elicitation for New Users in Collaborative Filtering
Wonbin Kweon, SeongKu Kang, Junyoung Hwang, Hwanjo Yu
WWW'20-short | ACM International World-Wide Web Conference
22. Multi-Modal Component Embedding for Fake News Detection
SeongKu Kang, Junyoung Hwang, Hwanjo Yu
IMCOM'20 | IEEE International Conf. Ubiquitous Information Management and Communication
23. Semi-Supervised Learning for Cross-Domain Recommendation to Cold-start Users
SeongKu Kang, Junyoung Hwang, Dongha Lee, Hwanjo Yu
CIKM'19 | ACM International Conference on Information and Knowledge Management
24. Densifying a Trust Network for Effective Collaborative Filtering
SeongKu Kang, Jemin Wang, Yeon-Chang Lee, Sang-Wook Kim
KDBC'17 | Korean DataBase Conference

Journals

1. Unbiased, Effective, and Efficient Distillation from Heterogeneous Models for Recommender Systems
SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu
ACM Transactions on Recommender Systems (2024.02)
2. Mitigating Viewpoint Sensitivity of Self-Supervised One-Class Classifiers
Hyunjun Ju, Dongha Lee, SeongKu Kang, Hwanjo Yu
Information Sciences (SCI) (2022.09)
3. Personalized Knowledge Distillation for Recommender System
SeongKu Kang, Dongha Lee, Wonbin Kweon, Hwanjo Yu
Knowledge-Based Systems (SCI) (2022.03)
4. Item-side Ranking Regularized Distillation for Recommender System
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu
Information Sciences (SCI) (2021.11)

PROFESSIONAL SERVICES

Program Committee/Reviewer

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD): 2023, 2024
- ACM International World-Wide Web Conference (WWW): 2024
- AAAI Conference on Artificial Intelligence (AAAI): 2024
- SIAM Conference on Data Mining (SDM): 2024

Journal Reviewer

- Knowledge-Based Systems: 2023, 2024
- Neural Computing and Applications: 2022

PATENTS

Issued

- Method and Apparatus for Improving the Accuracy of Speech Recognition Technology Based on Text Data Analysis for Deaf Students (Registration Number: 10-1988165-0000 | 2019.06.04 | South Korea)

Pending

- Distillation from Heterogeneous Models for Top-K Recommendation (2023)

- Molecular Clustering Device And Method Based on Deep Learning Technology Using Molecular Topology Information (2023)

REFERENCES

Prof. Jiawei Han (PostDoc advisor), UIUC

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Prof. Hwanjo Yu (PhD advisor), POSTECH

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Dr. Xing Xie (Internship mentor), MSRA

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Dr. Jianxun Lian (Internship mentor), MSRA

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